DEEP-DIVE ARCHITECTURAL & LOGIC AUDIT #04B: WEAPONS ARSENAL (PART 2 — BRAWLING, BALLISTIC & FLEXIBLE FAMILIES: `FIST`, `CLAW`, `BOW`, `NEEDLE`, `WHIP`)

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` / `SIM.PATCH_VERSION = '2.6.0'` / `HC.FINGERPRINT = '155:2f9c9dd7'`)
- **Target Audience**: Autonomous AI Software Engineers (**GPT** & **Gemini** Models)
- **Weapon Audit Series Plan (`04A–04E`)**:
  - `04A`: Group 1 — Core Bladed & Polearm Melee (`SWORD`, `BLADE`, `SPEAR`, `STAFF`)
  - **`04B` (This Document)**: Group 2 — **Brawling, Ballistic & Flexible (`FIST` หมัด/สนับมือ, `CLAW` กรงเล็บอสูร, `BOW` ธนู/หน้าไม้กล, `NEEDLE` เข็มบิน/เข็มฝังจุด/อาวุธลับ, `WHIP` แส้/โซ่/ด้ายปราณ)**
  - `04C`: Group 3 — Scholarly, Musical & Dharma Conduits (`FOCUS`, `ZITHER`, `BRUSH`, `ROSARY`)
  - `04D`: Group 4 — Grand Dao Tactical Implements (`FAN`, `ASTROLABE`, `ABACUS`)
  - `04E`: Part 5 — Holistic Weapon Ecosystem & Cross-System Catch-All Audit

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0. Hard Architectural Invariants for AI Implementers

1. **Immutable `human_core.js`**: Never modify `GAME_SOURCE/js/human_core.js` (`HC.FINGERPRINT === '155:2f9c9dd7'`). Remember that `h.skills` uses key `'FIST_PALM'` for unarmed/gauntlet proficiency while `gearType` is `'FIST'`, and newer weapon proficiencies (`CLAW`, `NEEDLE`, `WHIP`) route through `M.skillRead` / `M.skillAdd` without polluting `HC.validate(n)`!
2. **40-Script Load Order & Closed-Loop Conservation**: Preserve `<script>` order in `index.html`; all clue rewards and weapon repairs must obey world uniqueness and resource conservation.
3. **100% Emergent NPC Autonomy**: Golden Needle healers should naturally covet Divine Needles, Beast Tamers should favor Whips, Archers must actually kite when cornered, and Claw Shifters should never be penalized for wearing Claws.

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1. [WPN-FIST-01] Weapon Family #5: FIST (หมัด / สนับมือ / ปลอกแขนวัชระ)

1.1 Naming Collision Between Procedural FIST & CLAW (js/economy.js:583)!

- **Exact Code (economy.js:583, 590)**:

```
JavaScript
FIST: ['สนับมือเหล็กดำ', 'กรงเล็บพยัคฆ์', 'ปลอกแขนวัชระ', 'สนับมือหยกม่วง', 'กรงเล็บเหยี่ยวศึก'],
CLAW: ['กรงเล็บเหล็กพยัคฆ์', 'กรงเล็บเขี้ยวหมาป่า', ...],
```

- **Bug & Player/AI Confusion**:
  - **40% (2/5) of procedural FIST base names** (`'กรงเล็บพยัคฆ์'`, `'กรงเล็บเหยี่ยวศึก'`) literally name the item **"กรงเล็บ" (Claw)** even though `gearType === 'FIST'`!
  - Worse: if `PRIMORDIAL_SHIFTER` or `VENOM_GU.SCORPION_CLAW_EXECUTIONER` equips a `'กรงเล็บพยัคฆ์'` rolled from `REC_PROC_FIST`, every `CLAW`-typed technique suffers a **`×0.65` mismatch penalty**; and if an un-tagged item passes through `JD.MARTIAL.classify('WEAPON', nameTh)`, `'กรงเล็บ'` collides between `FIST` and `CLAW`!
- **Fix**: Replace the two `'กรงเล็บ...'` entries in `PROCEDURAL_POOLS.baseForms.FIST` with authentic gauntlet/bracer forms (`'สนับหมัดเหล็กไหล'`, `'ปลอกแขนพยัคฆ์คำราม'`).

1.2 Weakest Procedural Stat Budget (economy.js:689) & Unarmed Callus Gap

- **Stat Starvation in `typedBase` (economy.js:684–699)**:
  - `CLAW` gets `{ atk: +0.16*baseScale, crit: 3+0.6t, pen: 4+t }`; `BLADE` gets `{ atk: +0.14*baseScale, critDmg }`; `STAFF` gets `{ def: +0.16*baseScale, hp: +0.14*baseScale }`.
  - **`FIST` gets ONLY `{ cast: 4 + tier, crit: 1.5 + tier * 0.4 }` (`+0 atk`, `+0 def`, `+0 hp`, `+0 pen`)**—leaving `FIST` ~15% weaker than both `BLADE` and `CLAW`! Add `atk: baseScale * 0.10, def: baseScale * 0.08` to `typedBase.FIST`.
- **Bare-Handed Martial Artist Fallback**: When an NPC has **no weapon equipped** (`!n.equip.WEAPON`) and uses `FIST_PALM` + `BODY_REFINING` arts (e.g., *Eighteen Dragon-Subduing Palms* or `CROWD_ROAR_BRAWLER`), grant a bounded **Iron-Palm Callus Baseline** (`atk += BODY_REFINING * 1.4 + FIST_PALM * 1.2`) in

`ST.get` / `ST.explain` so bare-knuckle masters aren't helpless if their gauntlets break mid-expedition!

1.3 `LEG_WPN_WRATHFUL_TIGER_GAUNTLETS` (สนับหมัดพยัคฆ์พิโรธ) Bypasses `healBlock` & Wastes ICD (`visitors_clues.js:835–844`)

- Sets `wpnProcCds[eid] = W.t + icd` unconditionally on line 836 even at `100%` HP in a 1v1 duel (where neither `hpLow` nor `hasAllies` is true!), and calls raw `COMBAT.setHp(...)` on line 841 instead of `COMBAT.healUnit`, **bypassing `healBlock` / `stygianFreeze`**! Route through `COMBAT.healUnit` and gate ICD on actual activation.

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2. `[WPN-CLAW-02]` Weapon Family #6: `CLAW` (กรงเล็บอสูร / กรงเล็บพิษ)

2.1 ZERO Clue Templates in `VC.CLUE_TEMPLATES` for `LEG_WPN_TAOTIE_DEVOURING_CLAW`!

- **Locations**: `js/visitors_clues.js:353–376, 475` & `test/visitors_clues_weapons_suite.js:26–28, 254–255`
- **Verified Reachability Gap**:
  - `LEG_WPN_TAOTIE_DEVOURING_CLAW` (กรงเล็บเทพอสูรเทาเที่ยกลืนสวรรค์) stamps `obtainSources: ['CLUE_QUEST', 'SECRET_REALM', 'RELIC_INHERITANCE']`, **yet `VC.CLUE_TEMPLATES` (`60` templates) contains ZERO clue chain for `TAOTIE_DEVOURING_CLAW`** (or any of the 6 `v2.0.0` weapon families)! Visitors and tavern rumors can never spawn a quest for it!
  - **Fix**: Author `CLUE_LEG_WPN_TAOTIE_DEVOURING_CLAW` (แท่นสังเวยอสูรเทาเที่ยในหุบเหวบรรพกาล — `guardian_combat` vs Taotie Remnant + personal `CLAW` / `physiology.explosiveness >= 60` trial)!

2.2 Legacy Claw Techniques Penalize `CLAW` Weapons (`×0.65` Mismatch) & Missing Pet/Subpath Hooks

1. **Weapon-List Omission in `data_tech.js` / `data_martial.js`**: Pre-`v2.0.0` claw/talon arts (`กรงเล็บ...`, `DEVOURING_FIEND`, `TIGER_SOVEREIGN`, `WHITE_TIGER`) list `weaponTypes: ['FIST']` without `'CLAW'`, punishing actual `CLAW` weapons with `0.65×` damage! Append `'CLAW'` to every technique whose `nameTh` or theme involves claws/talons (`กรงเล็บ|ขย้ำ|ฉีกกระชาก`).

2. **Missing `'CLAW'` in `M.gearSuitability` Pet Synergy (`js/martial.js:569`)**: Add `'CLAW'` alongside `'FIST', 'BLADE', 'SPEAR', 'STAFF'` for `FEROCIOUS_ASSAULT` beast-pack synergy!
3. **Inert `EFX_V200_ATAVISM_CLAW_REND` When Equipped & Unchecked Lifesteal**:
    - `EFX_V200_ATAVISM_CLAW_REND` is handled in `ECO.executeItemEffectById` (consumables) rather than on-hit weapon procs in `combat.js`! Wire an on-hit bleed/armor-shred proc in `runItemEffectProcs`.
    - In `visitors_clues.js:886–889`, route `TAOTIE_DEVOURING_CLAW`'s `leechRatio: 0.08` through `COMBAT.healUnit` (respecting `healBlock` and halving leech against bloodless stone/metal constructs!).

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3. `[WPN-BOW-03]` Weapon Family #7: `BOW` (ธนูศึก / เกาทัณฑ์ / หน้าไม้กล)

3.1 `LEG_WPN_CONDOR_METEOR_BOW` Inheritance Trial Forgot `skill: 'BOW'` (`visitors_clues.js:1487–1498`)!

- **Exact Code (`visitors_clues.js:1487–1498`)**:

```JavaScript
{
  nodeId: 'shoot_wind_vane_target',
  type: 'skill_trial',
  titleTh: 'ยิงศรฝ่าลมกรดตัดห่วงทองคำสองวงพร้อมกัน',
  locationId: 'SRC110_MW_ENV_05_MINERAL_DESERT',
  scope: 'personal',
  statPath: 'intellect.perceptiveness',
  minVal: 58, // <-- MISSING skill: 'BOW'!
}
```

- **Contextual Absurdity**: Every other Legendary Weapon `skill_trial` checks its weapon skill (`SWORD`, `BLADE`, `SPEAR`, `ZITHER`, `BRUSH`, `WHIP`), **but `CONDOR_METEOR_BOW`'s twin-ring archery shot checks ONLY `intellect.perceptiveness`**, allowing a scholar with **`BOW === 0`** to out-shoot the sect's master archers and claim the unique Meteor Warbow! Add `skill: 'BOW'`!
- Also fix `VC.benchmarkWeaponBalance` (`visitors_clues.js:2114`) where `BOW` is benchmarked against `'MP_SWORD_5'` instead of `'MP_BOW_5'`.

3.2 Broken Kiting Disengage Traps Archers in `BOW_CAUGHT_IN_MELEE` (`js/combat.js:890`)

- When a melee enemy closes to `gap <= 1.05`, `BOW` suffers `BOW_CAUGHT_IN_MELEE` (`×0.92` dmg, `-6` acc). When the archer AI casts an `evade`/`retreat` skill to leap back to kiting range, `useSkill` (`combat.js:890`) `return`s **before** running the `isRetreat` coordinate pushback (`line 961`), trapping the archer at `gap <= 1.05` permanently! Fixing `useSkill` retreat repositioning restores authentic `BOW` kiting!

3.3 Mechanical Crossbow (`หน้าไม้กล` / `CELESTIAL_BALLISTA`) vs Warbow Sub-Identity

- Differentiate `visualSubtype === 'CROSSBOW'` / `MECH_PUPPET.CELESTIAL_BALLISTA` (`+16 pen` armor-piercing bolts, ignores `30%` wind/weather accuracy penalty, slightly slower `cast`) from swift recurve bows (`STAR_ARCHER` / `SHADOW_ARCHER`).

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4. `[WPN-NEEDLE-04]` Weapon Family #8: `NEEDLE` (`เข็มบิน` / `เข็มพิษ` / `เข็มทองฝังจุด`)

4.1 Absurd Profile Bug: `GOLDEN_NEEDLE_SAINT` (สายปราชญ์เข็มทอง) Treats `NEEDLE` as an Unfamiliar Weapon (`0.72×` Suitability)!

- **Exact Code (`js/martial.js:82` & `js/visitors_clues.js:383`)**:

```
JavaScript
GOLDEN_NEEDLE_SAINT: [{dexterity:26,spiritualFoundation:24,reflex:20},
{MEDICINE:.38,ALCHEMY:.24,FOCUS:.22}, 'steady', ['FOCUS','SWORD'],
['ROBE','LIGHT']],
```

- **Chain-Reaction Failure**:
  1. `M.PROFILES.GOLDEN_NEEDLE_SAINT[3]` lists **`['FOCUS', 'SWORD']`**—omitting `'NEEDLE'`! When a *Golden Needle Saint* evaluates any `NEEDLE` weapon in `M.gearSuitability` (`martial.js:534`), it falls into `else { fit = 0.72; reasons.push('อาวุธไม่ถนัด'); }` and refuses to equip Needles!

2. Even `LEG_WPN_BAIHUA_SOUL_NEEDLE` (`visitors_clues.js:383`) omitted `'GOLDEN_NEEDLE_SAINT'` and `'CONCEALED_WEAPON_STRIKER'` (`NINPO_SHADOW`) from `preferredSubs`!
3. Pre-`v2.0.0` needle techniques in `data_tech.js` / `data_martial.js` still lock `weaponTypes: ['FOCUS']`, penalizing `NEEDLE` wielders by `×0.65`!

- **Fixes**:
  - Set `GOLDEN_NEEDLE_SAINT` preferred weapons to `['NEEDLE', 'FOCUS', 'SWORD']` (and add `NEEDLE: 0.32` life/martial scaling + `M.SUB_EQUIP_BONUSES.GOLDEN_NEEDLE_SAINT`), add `'NEEDLE'` to `CONCEALED_WEAPON_STRIKER` and `VENOM_GU_SOVEREIGN`, and append `'NEEDLE'` to all acupuncture/flying-needle techniques!
  - Author `CLUE_LEG_WPN_BAIHUA_SOUL_NEEDLE` in `VC.CLUE_TEMPLATES` (ศาลาหมอยาเร้นลับในหุบเขาร้อยบุปผา — `MEDICINE`/`POISON` + `NEEDLE` acupuncture trial).
  - Verify `meridianSeal` (`EFX_LEG_BAIHUA_SOUL_NEEDLE` `targetStatus` at `visitors_clues.js:391`) applies concrete `-35%` Qi regen & `-18` `cast` speed in `STATS`/`COMBAT`, and grant `+15%` `A.heal` meridian-treatment speed when the sect physician equips a T4+ `NEEDLE`!

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5. `[WPN-WHIP-05]` Weapon Family #9: `WHIP` (แส้ / โซ่ตรวน / ด้ายไหมปราณ)

5.1 Orphaned Renamed Subpath IDs in `M.SUB_EQUIP_BONUSES` & `LEG_WPN_TIANCAN_FATE_WHIP`!

- **Locations**: `js/martial.js:476–478`, `js/visitors_clues.js:315`
- **Verified Stale IDs**:
  - `M.SUB_EQUIP_BONUSES` keys on `'SPIRIT_THREAD_PUPPET'` (aliased away!) while `LEG_WPN_TIANCAN_FATE_WHIP.preferredSubs` lists `'SPIRIT_THREAD_PUPPET'`, `'PACK_COMMAND_HUNT'`, `'THOUSAND_THREAD_PUPPETEER'`, and `'RED_THREAD_ARBITER'`—missing **all 6 real `SOUL_PUPPETRY` subpaths (`PHANTOM_SILK_PUPPETEER`, `FATE_STRING_CONDUCTOR`, etc.)**, **`SPIRIT_HUSBANDRY.IRON_THORN_VINE_WEAVER` (สายเถาวัลย์หนามเหล็ก)**, **`KARMA_SAMSARA.NETHER_RIVER_REAPER` (ยมทูตเคียวโซ่)**, and **`MECH_PUPPET.JADE_SPIDER_WEAVER`**!

5.2 Procedural `WHIP` Stat Skew Hurts Magical Thread/Karma Weavers (`js/economy.js:669, 693`)

- In `ECO.rollProceduralGear` (`economy.js:669`), `isFocus` excludes `'WHIP'`, and `typedBase.WHIP` (`line 693`) adds `{ atk: 0.10*baseScale, acc, pen }` with **`+0 matk`**!
- Yet over half of `WHIP` users (`KARMA_SAMSARA.FATE_SEVERING_THREAD`, `SOUL_PUPPETRY.PHANTOM_SILK_PUPPETEER`, `HEART_SOUL` thread weavers) are **`matk`-primary** (compare `LEG_WPN_TIANCAN_FATE_WHIP` which rightly has balanced `atk: 690, matk: 620`!).
- **Fix**: Give procedural `WHIP` (like `NEEDLE` and `ABACUS`) balanced hybrid scaling (`atk: 0.40*baseScale, matk: 0.40*baseScale` + `typedBase { atk: 0.08*baseScale, matk: 0.08*baseScale, acc: 0.14*baseScale, pen: 3 + tier }`), and grant `+10%` wild-beast taming control in `wildlife.js` when wielding a `WHIP`!

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6. Implementation & Verification Checklist for GPT / Gemini (`04B`)

- ☐ **`FIST`**: Remove `'กรงเล็บ...'` names from `PROCEDURAL_POOLS.baseForms.FIST`; buff `typedBase.FIST` (`+atk/+def`); add bare-handed `FIST_PALM` + `BODY_REFINING` baseline when unequipped; route `WRATHFUL_TIGER_GAUNTLETS` heal through `COMBAT.healUnit` and conditional ICD.
- ☐ **`CLAW`**: Add `CLUE_LEG_WPN_TAOTIE_DEVOURING_CLAW` to `VC.CLUE_TEMPLATES`; add `'CLAW'` to legacy claw techniques & `M.gearSuitability` `FEROCIOUS_ASSAULT` pet check; wire `EFX_V200_ATAVISM_CLAW_REND` in combat `runItemEffectProcs` and route `TAOTIE` leech through `COMBAT.healUnit`.
- ☐ **`BOW`**: Add missing `skill: 'BOW'` to `CLUE_LEG_WPN_CONDOR_METEOR_BOW` Step 2 (`visitors_clues.js:1493`); fix `benchmarkWeaponBalance 'MP_BOW_5'`; fix `useSkill` retreat movement so cornered archers actually kite out of `BOW_CAUGHT_IN_MELEE`.
- ☐ **`NEEDLE`**: Add `'NEEDLE'` as #1 preferred weapon on `GOLDEN_NEEDLE_SAINT`, `CONCEALED_WEAPON_STRIKER`, and `VENOM_GU_SOVEREIGN`; add `CLUE_LEG_WPN_BAIHUA_SOUL_NEEDLE`; wire `meridianSeal` status & infirmary acupuncture bonus; add `'NEEDLE'` to legacy needle techniques.
- ☐ **`WHIP`**: Update `M.SUB_EQUIP_BONUSES` & `TIANCAN_FATE_WHIP.preferredSubs` with live `SOUL_PUPPETRY`, `IRON_THORN_VINE_WEAVER`, `NETHER_RIVER_REAPER`, and `JADE_SPIDER_WEAVER` IDs; make procedural `WHIP` hybrid `atk`/`matk`; add `+10%` beast-subdue bonus.